Bluestock Mutual Fund Analytics Capstone

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Executive Summary

The Bluestock Mutual Fund Analytics Capstone project was developed to perform end-to-end analysis of mutual fund performance, investor behaviour, and industry trends using data analytics and business intelligence tools. The primary objective of the project was to transform raw mutual fund datasets into actionable insights through data cleaning, exploratory analysis, financial performance evaluation, advanced risk analytics, and interactive dashboard visualization.

The project utilized multiple datasets, including mutual fund NAV history, assets under management (AUM), scheme performance metrics, and investor transaction records. An ETL pipeline was designed to ingest, clean, validate, and transform the raw data into analysis-ready datasets. The cleaned data was then used for exploratory data analysis (EDA) to understand fund characteristics, investor demographics, transaction patterns, and market trends.

Several performance analytics metrics were computed to evaluate mutual fund schemes, including daily returns, CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, and benchmark comparisons. Advanced analytics techniques such as Historical Value at Risk (VR), Conditional Value at Risk (CVR), Rolling Sharpe Ratio analysis, investor cohort analysis, SIP continuity analysis, and a rule-based fund recommendation system were also implemented to provide deeper insights into risk and investor behaviour.

An interactive Power BI dashboard was developed with four analytical pages: Industry Overview, Fund Performance, Investor Analytics, and SIP & Market Trends. The dashboard enables users to explore key performance indicators, compare funds, analyse investor trends, and monitor market-related metrics through interactive visualizations and filters.

The project successfully demonstrates how data engineering, financial analytics, and business intelligence can be integrated to support data-driven investment decisions. Key outcomes include identification of top-performing funds based on risk-adjusted returns, detection of at-risk SIP investors, analysis of investor cohorts, and creation of a comprehensive dashboard for mutual fund performance monitoring. The resulting analytics framework provides a scalable foundation for future enhancements such as live NAV integration, portfolio optimization, and automated reporting solutions.

Objectives

The primary objectives of the Bluestock Mutual Fund Analytics Capstone project were:

1. Build a robust ETL pipeline to ingest, clean, transform, and prepare mutual fund datasets for analysis.
2. Perform Exploratory Data Analysis (EDA) to identify patterns, trends, and anomalies in mutual fund and investor data.
3. Compute key financial performance metrics including CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, and benchmark comparisons.
4. Design and develop an interactive Power BI dashboard to visualize industry trends, fund performance, investor analytics, and SIP market insights.
5. Perform advanced analytics such as Value at Risk (VR), Conditional Value at Risk (CVR), Rolling Sharpe Ratio analysis, investor cohort analysis, and SIP continuity analysis.
6. Build a simple fund recommendation system that suggests suitable mutual funds based on investor risk appetite.
7. Generate actionable insights and recommendations to support data-driven investment decision-making.

Data Sources

The project utilized multiple datasets covering mutual fund performance, asset management statistics, and investor behaviour. These datasets were processed and integrated to support comprehensive analysis.

Dataset	**Purpose**
NAV History	Calculation of daily returns, CAGR, drawdowns, and risk metrics
AUM Data	Analysis of asset size, fund house market share, and industry trends
Scheme Performance	Evaluation of returns, risk-adjusted performance, Alpha, Beta, Sharpe Ratio, and Sortino Ratio
Investor Transactions	Analysis of investor behavior, SIP patterns, cohort analysis, and transaction trends

ETL Pipeline Design

The ETL (Extract, Transform, Load) process was designed to ensure data quality and consistency before analysis.

Data Ingestion

Raw CSV datasets were imported into Python using Pandas and validated for completeness and schema consistency.

Data Cleaning

Missing values, duplicate records, inconsistent formats, and incorrect data types were identified and corrected. Date fields were standardized and numerical fields were validated.

Data Transformation

Derived metrics such as daily returns, annualized performance measures, and investor analytics features were generated. Relevant datasets were merged and prepared for downstream analytics.

Data Export

The cleaned and transformed datasets were exported to the processed data layer for further analysis and dashboard development.

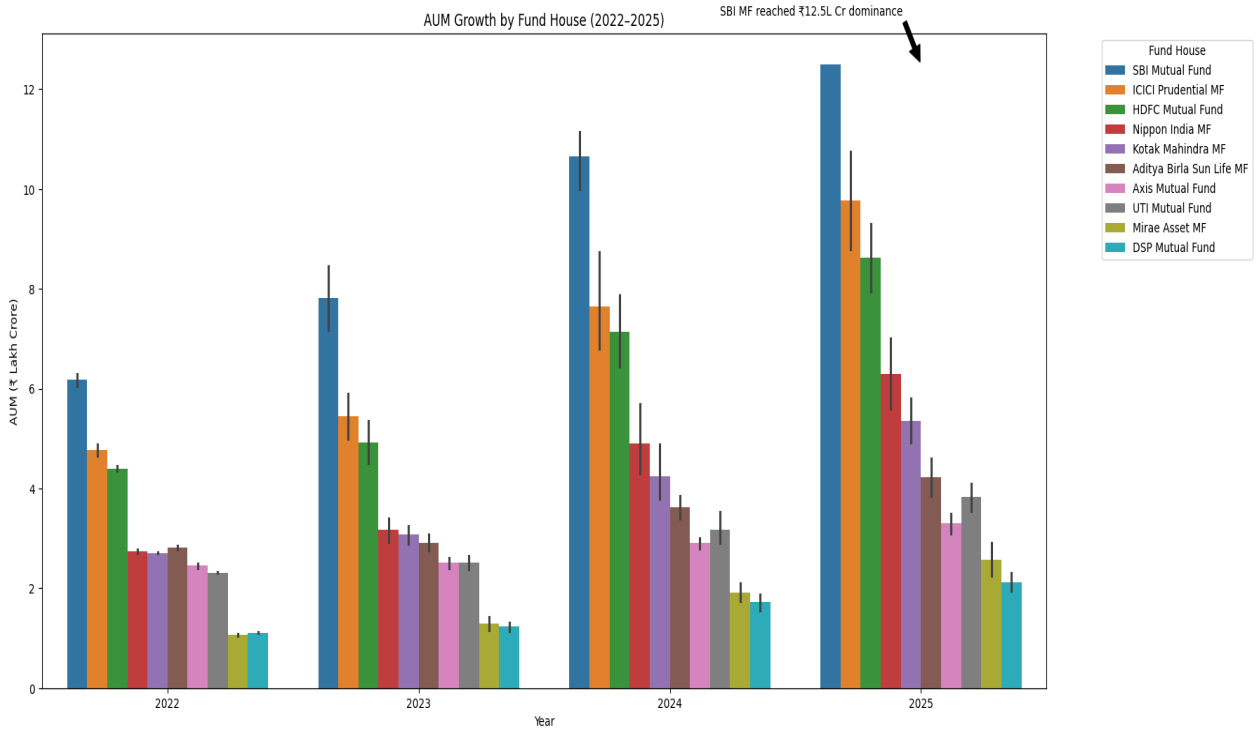
ETL Workflow:

Raw Data
↓
Data Cleaning
↓
Data Transformation
↓
Processed CSV Files
↓
Analytics & Dashboard Development

EDA Findings

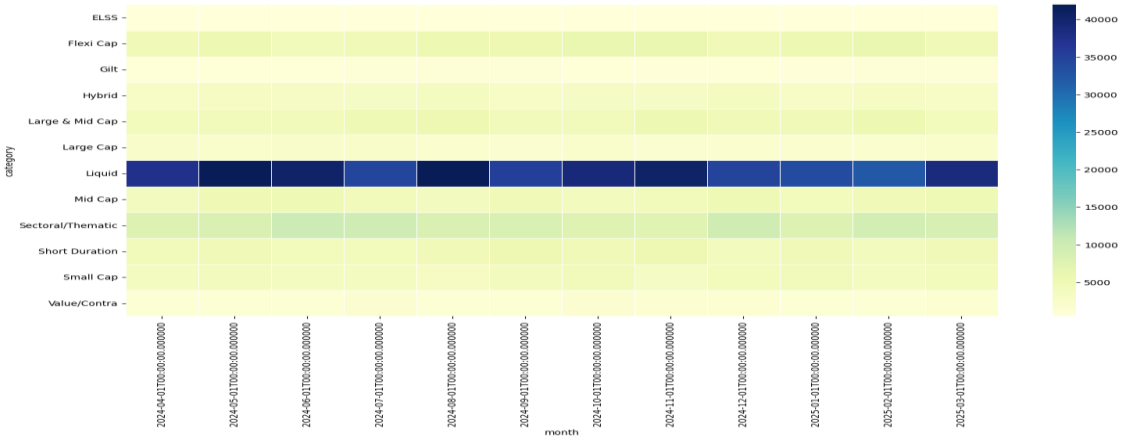
AUM Distribution Analysis

The AUM distribution analysis revealed significant concentration among a small number of fund houses. Large asset managers controlled a substantial share of total industry assets, indicating a highly competitive but concentrated market structure.



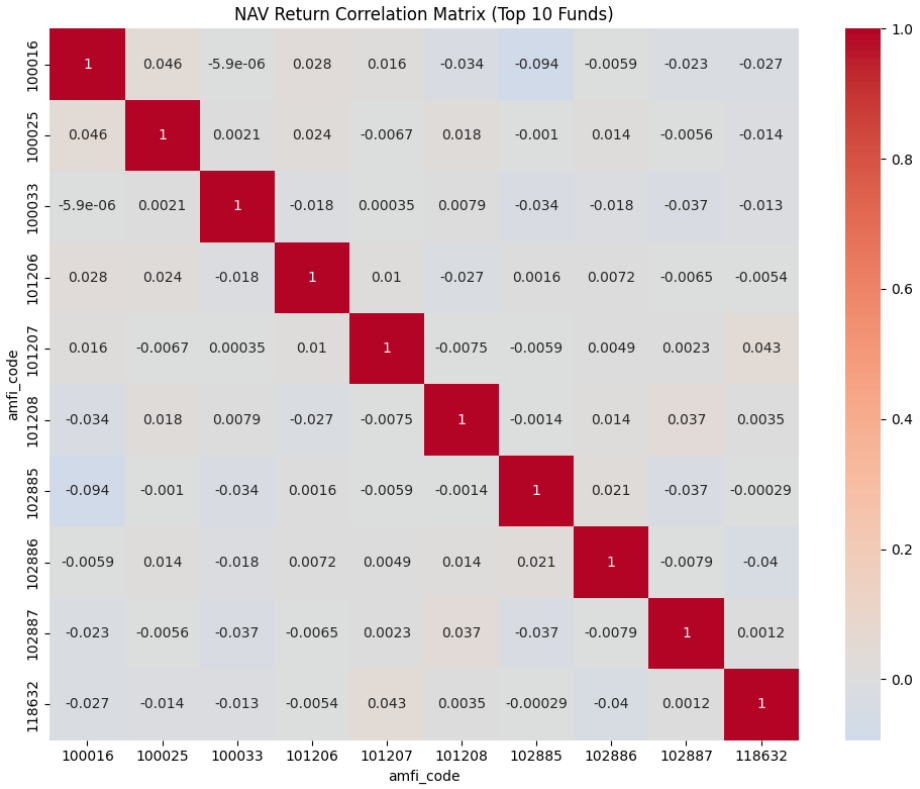
Fund Category Distribution

The category-wise analysis showed that equity-oriented schemes formed the largest segment of the dataset. Hybrid and debt categories contributed a smaller share but provided diversification opportunities for investors.



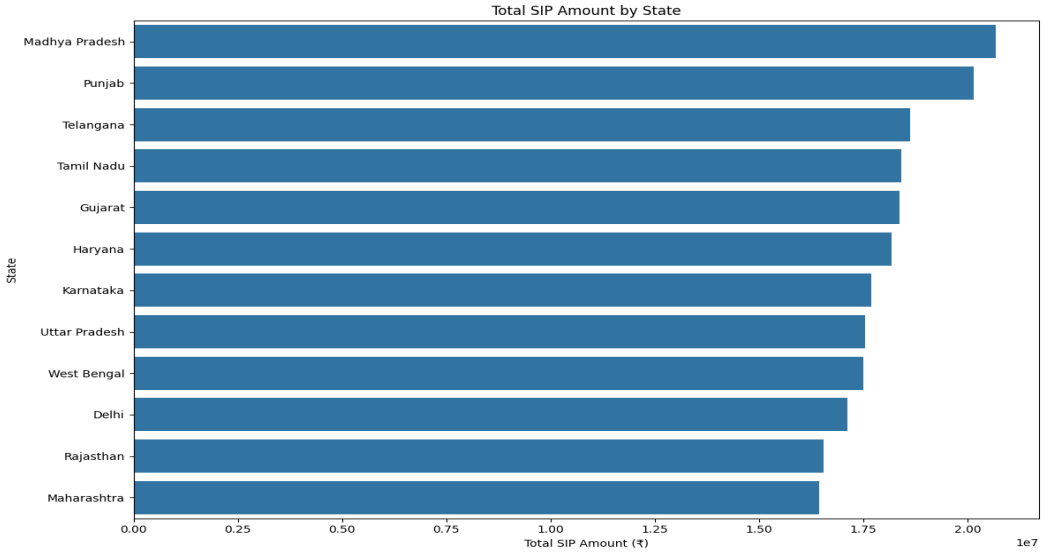
Return Correlation Analysis

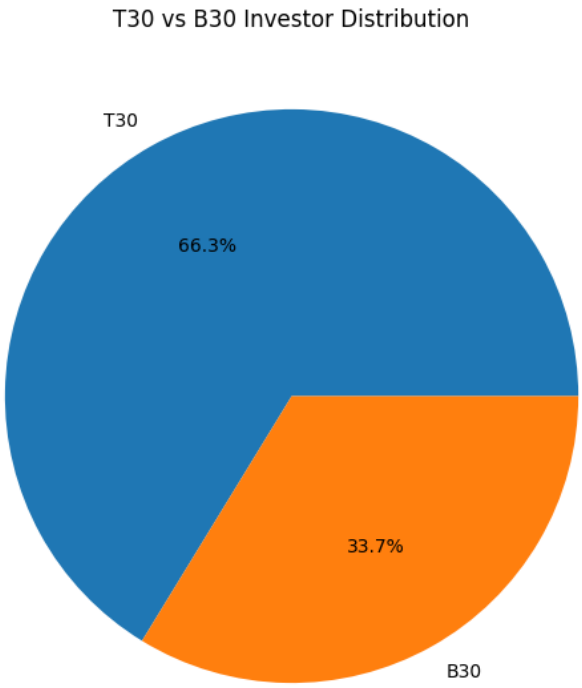
The correlation heat map highlighted relationships between scheme returns. Several funds exhibited strong positive correlations, indicating similar market exposure and investment strategies.

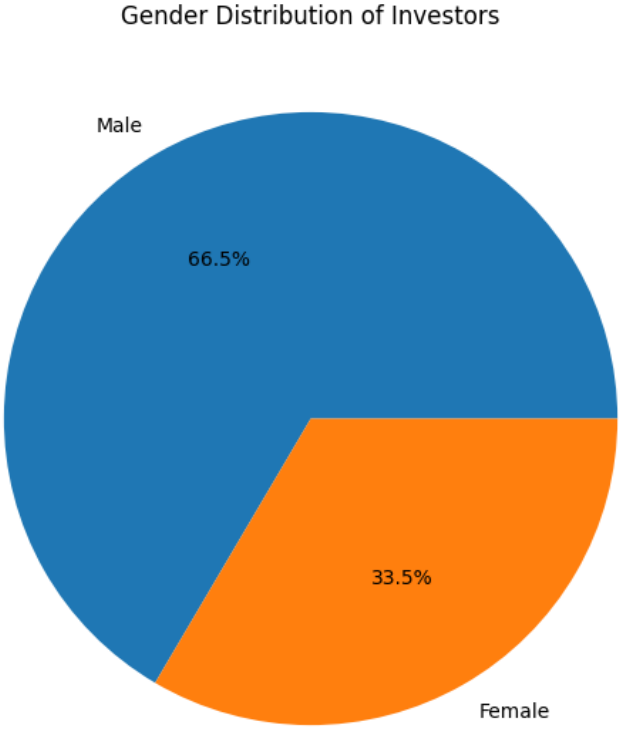


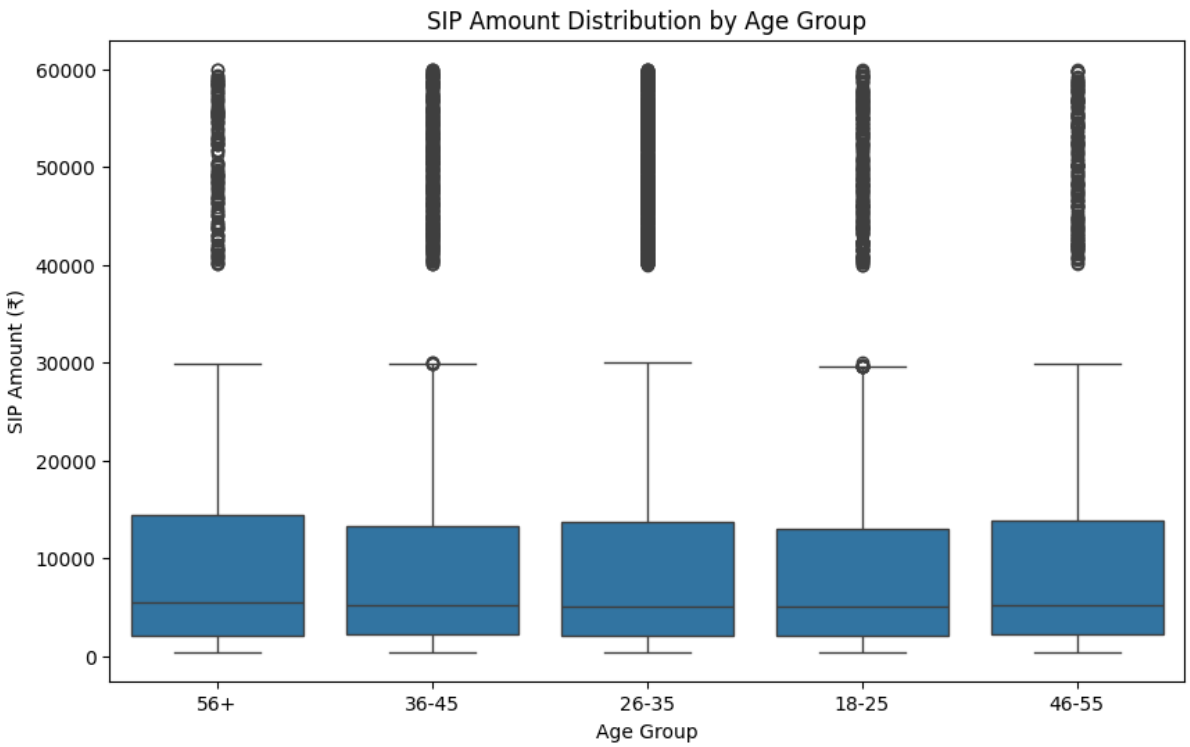
Investor Demographics Analysis

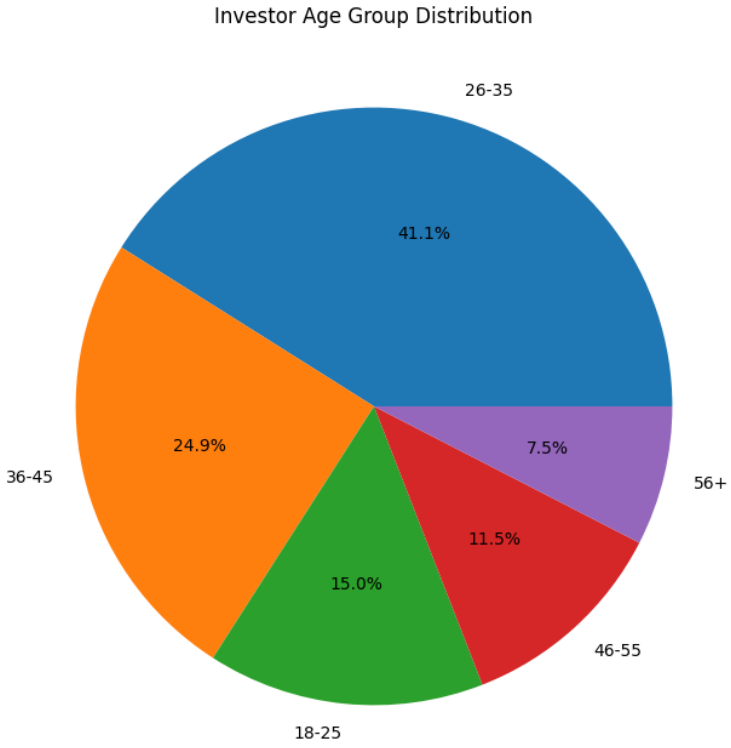
Investor transaction data indicated varying participation levels across age groups and geographic regions. Certain age groups demonstrated higher average investment amounts and stronger SIP adoption rates.




T30 vs B30 Investor Distribution
T30
66.3%
33.7%
B30


Gender Distribution of Investors
Male
66.5%
33.5%
Female


SIP Amount Distribution by Age Group
SIP Amount (₹)
60000
50000
40000
30000
20000
10000
0
56+
36-45
26-35
18-25
46-55
Age Group


Investor Age Group Distribution
26-35
41.1%
24.9%
36-45
7.5%
56+
11.5%
15.0%
46-55
18-25

Performance Analytics

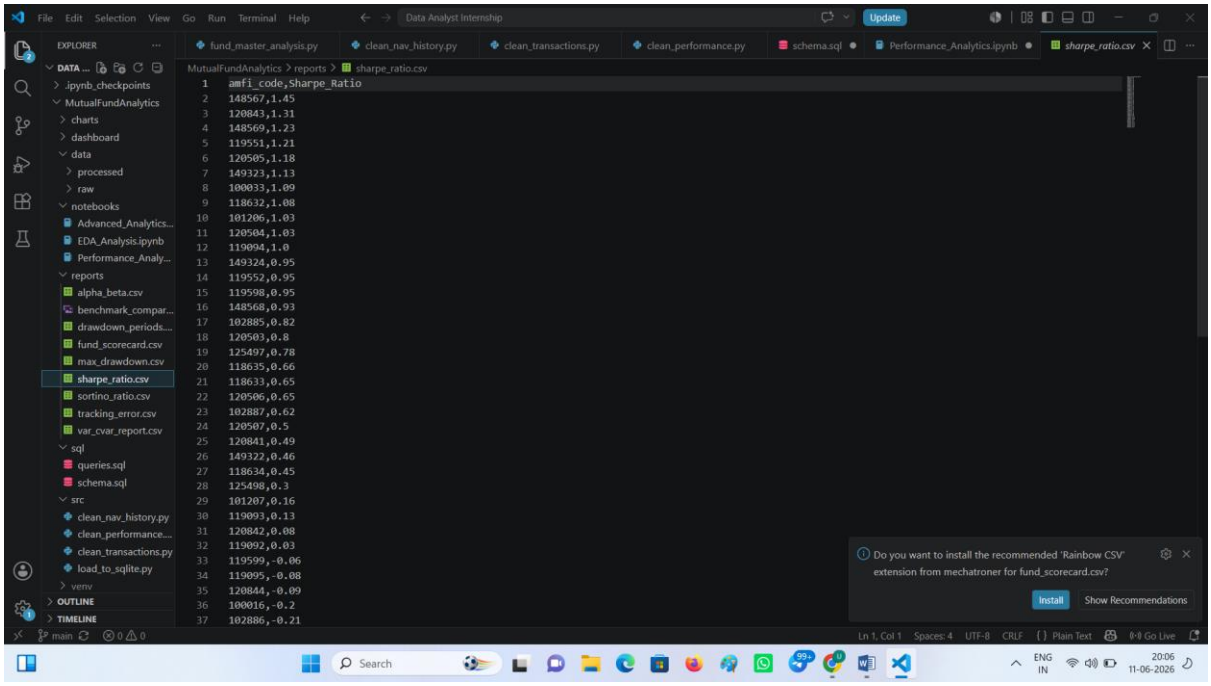
(Insert screenshots from fund_scorecard.csv, alpha_beta.csv, benchmark_comparison.png)

CAGR Analysis

Compound Annual Growth Rate (CAGR) was calculated for multiple investment horizons to evaluate long-term fund growth performance. Several schemes demonstrated consistent returns over extended periods.

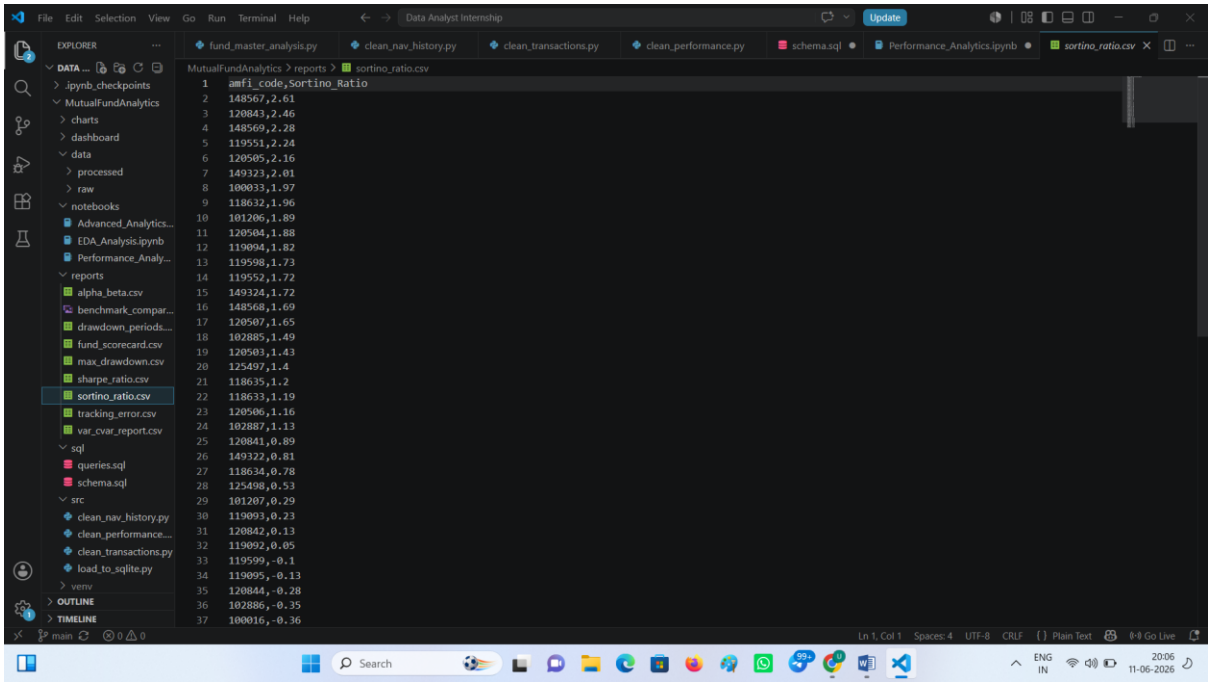
Sharpe Ratio Analysis

Sharpe Ratio analysis identified funds delivering superior risk-adjusted returns. Higher Sharpe Ratios indicated more efficient compensation for the risk undertaken by investors.



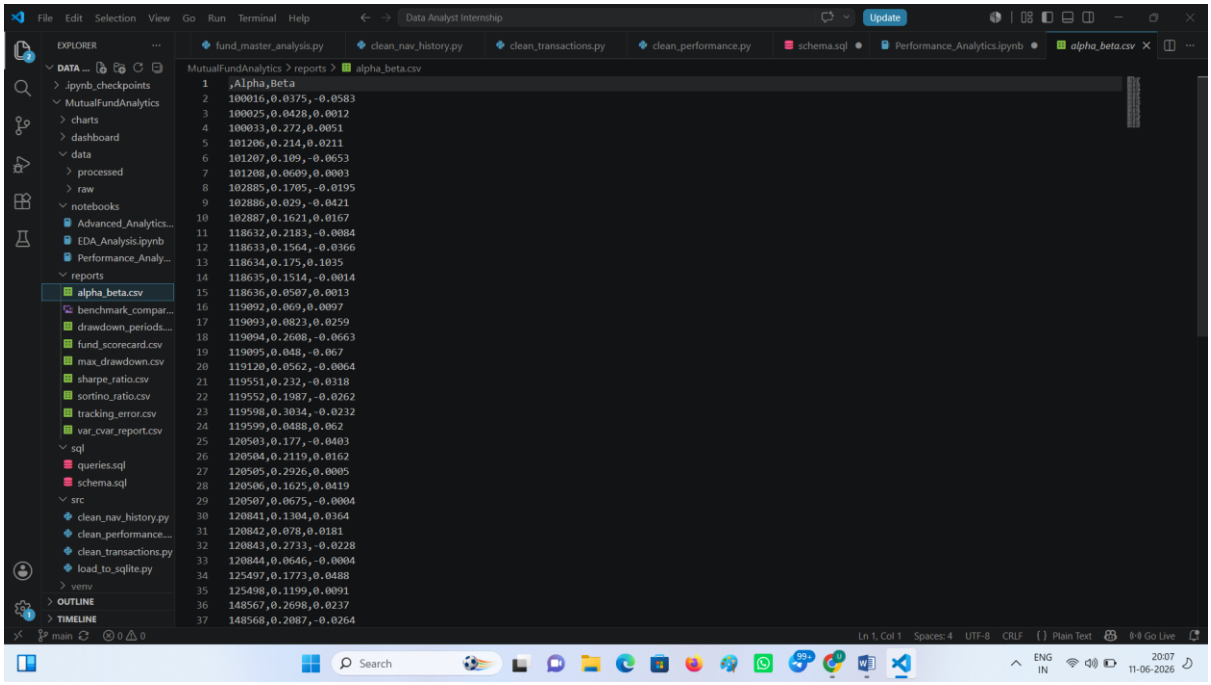
Sortino Ratio Analysis

Sortino Ratio measurements focused specifically on downside risk and highlighted funds with stronger downside protection characteristics.



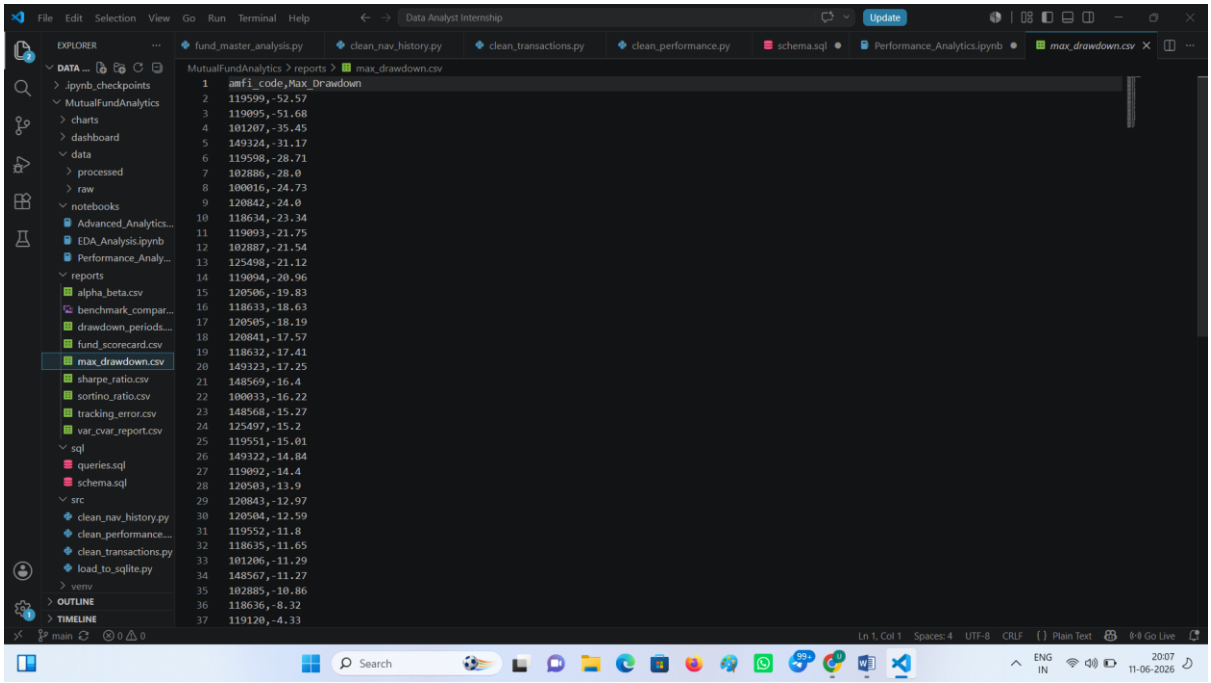
Alpha and Beta Analysis

Alpha values measured excess returns generated relative to benchmark expectations, while Beta values quantified market sensitivity. High-alpha funds demonstrated superior active management performance.

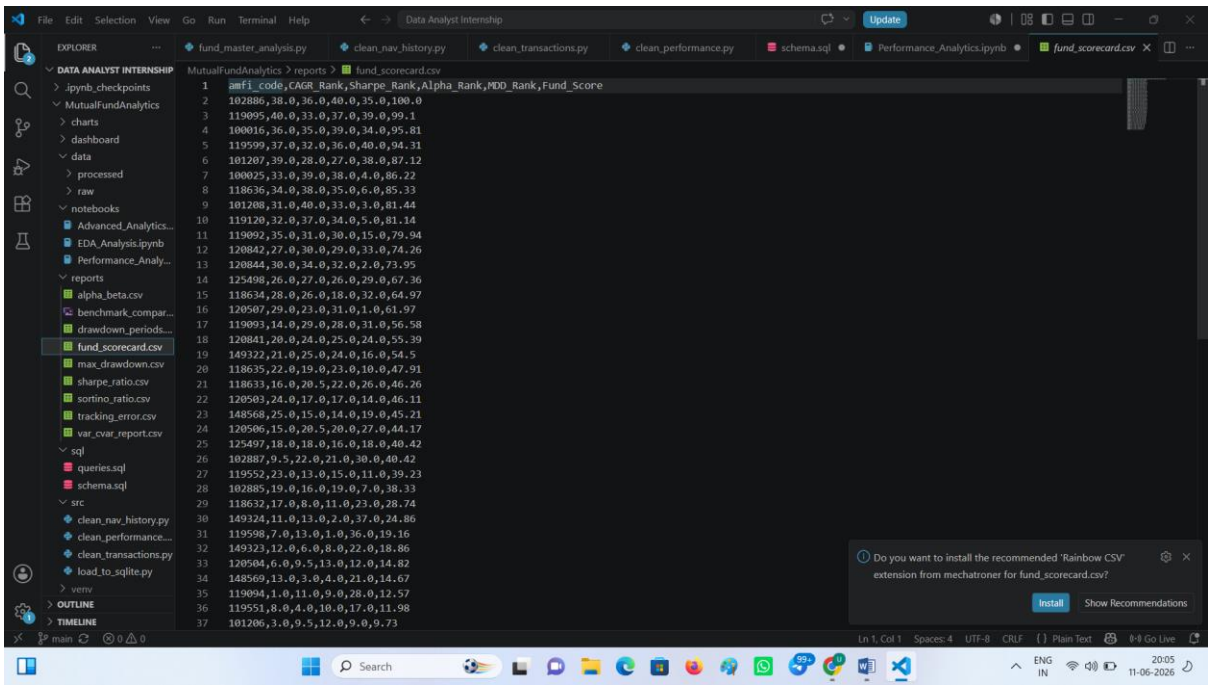


Maximum Drawdown Analysis

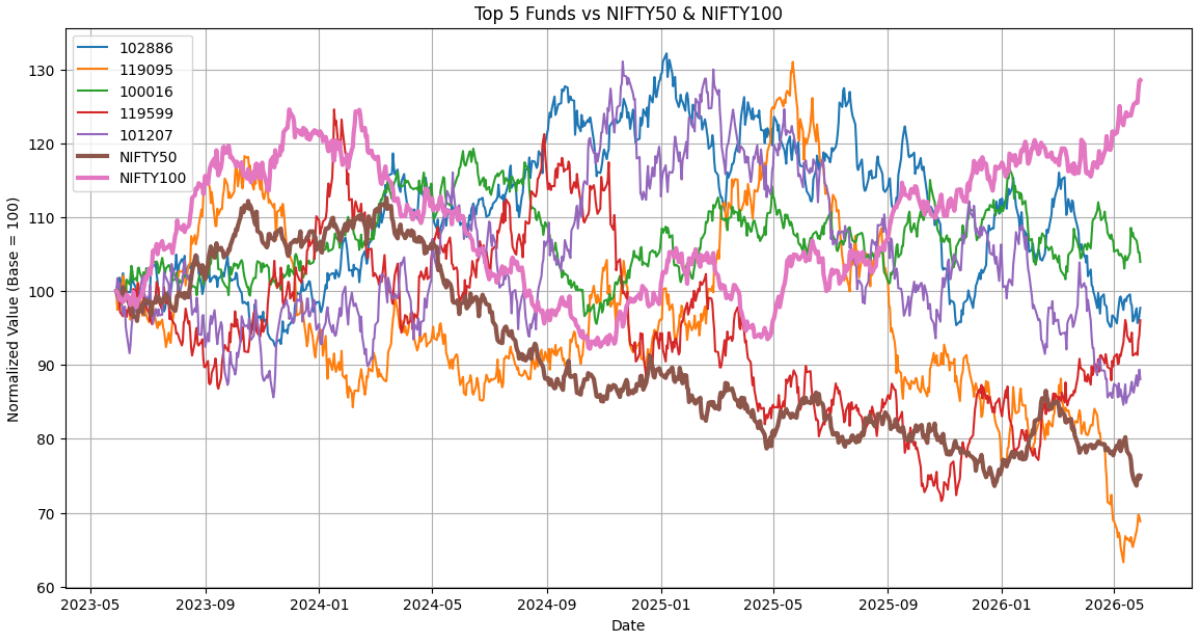
Maximum Drawdown calculations measured the largest historical declines experienced by each fund, providing insight into downside risk exposure during adverse market conditions.

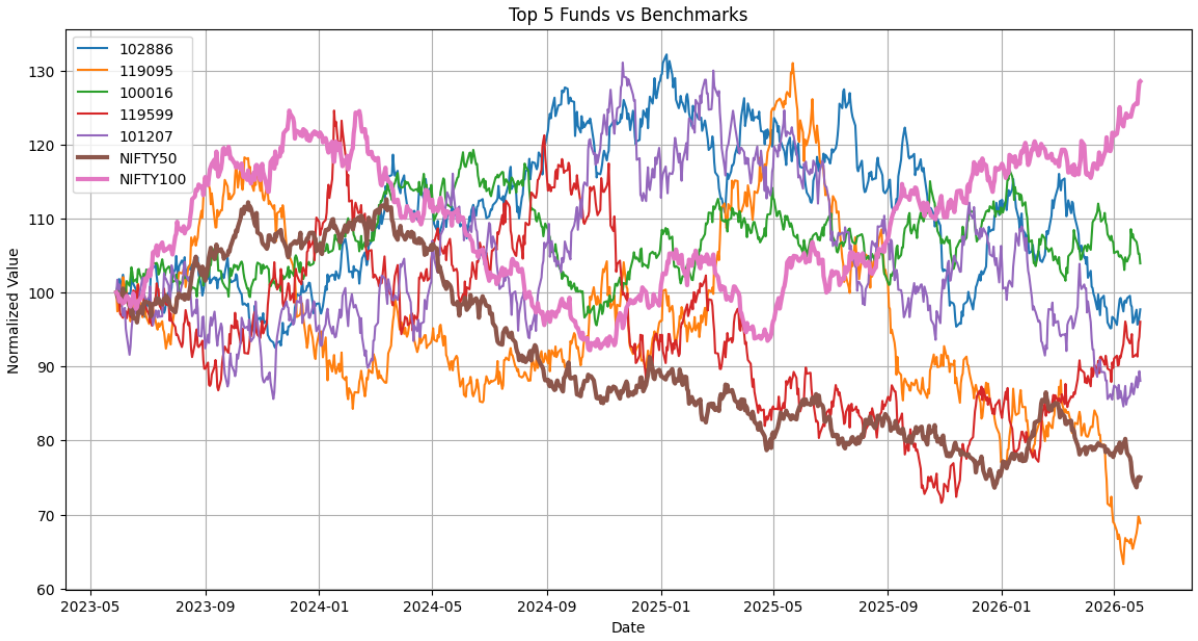


```
amfi_code,Max_Drawdown
119599,-52.57
119095,-51.68
101207,-35.45
149324,-31.17
119598,-28.71
102886,-28.0
100016,-24.73
120842,-24.0
118634,-23.34
119093,-21.75
102887,-21.54
125498,-21.12
119094,-20.96
120506,-19.83
118633,-18.63
120505,-18.19
120841,-17.57
118632,-17.41
149323,-17.25
148569,-16.4
100033,-16.22
148568,-15.27
125497,-15.2
119551,-15.01
149322,-14.84
119092,-14.4
120503,-13.9
120843,-12.97
120504,-12.59
119552,-11.8
118635,-11.65
101206,-11.29
148567,-11.27
102885,-10.86
118636,-8.32
119120,-4.33
```



```
amfi_code,CAGR_Rank,Sharpe_Rank,Alpha_Rank,MDD_Rank,Fund_Score
102886,38.0,36.0,40.0,35.0,100.0
119095,40.0,33.0,37.0,39.0,99.1
100016,36.0,35.0,39.0,34.0,95.81
119599,37.0,32.0,36.0,40.0,94.31
101207,39.0,28.0,27.0,38.0,87.12
100025,33.0,39.0,38.0,4.0,86.22
118636,34.0,38.0,35.0,6.0,85.33
101208,31.0,40.0,33.0,3.0,81.44
119120,32.0,37.0,34.0,5.0,81.14
119092,35.0,31.0,30.0,15.0,79.94
120842,27.0,30.0,29.0,33.0,74.26
120844,30.0,34.0,32.0,2.0,73.95
125498,26.0,27.0,26.0,29.0,67.36
118634,28.0,26.0,18.0,32.0,64.97
120507,29.0,23.0,31.0,1.0,61.97
119093,14.0,29.0,28.0,31.0,56.58
120841,20.0,24.0,25.0,24.0,55.39
149322,21.0,25.0,24.0,16.0,54.5
118635,22.0,19.0,23.0,10.0,47.91
118633,16.0,20.5,22.0,26.0,46.26
120503,24.0,17.0,17.0,14.0,46.11
148568,25.0,15.0,14.0,19.0,45.21
120506,15.0,20.5,20.0,27.0,44.17
125497,18.0,18.0,16.0,18.0,40.42
102887,9.5,22.0,21.0,30.0,40.42
119552,23.0,13.0,15.0,11.0,39.23
102885,19.0,16.0,19.0,7.0,38.33
118632,17.0,8.0,11.0,23.0,28.74
149324,11.0,13.0,2.0,37.0,24.86
119598,7.0,13.0,1.0,36.0,19.16
149323,12.0,6.0,8.0,22.0,18.86
120504,6.0,9.5,13.0,12.0,14.82
148569,13.0,3.0,4.0,21.0,14.67
119094,1.0,11.0,9.0,28.0,12.57
119551,8.0,4.0,10.0,17.0,11.98
101206,3.0,9.5,12.0,9.0,9.73
```


Top 5 Funds vs NIFTY50 & NIFTY100
102886
119095
100016
119599
101207
NIFTY50
NIFTY100
Normalized Value (Base = 100)
Date
60
70
80
90
100
110
120
130
2023-05
2023-09
2024-01
2024-05
2024-09
2025-01
2025-05
2025-09
2026-01
2026-05


Top 5 Funds vs Benchmarks
102886
119095
100016
119599
101207
NIFTY50
NIFTY100
Normalized Value
Date
60
70
80
90
100
110
120
130
2023-05
2023-09
2024-01
2024-05
2024-09
2025-01
2025-05
2025-09
2026-01
2026-05